

NeuroPolicy: Risk-Aware Offline RL with Calibrated Off-Policy Evaluation for Motor-Imagery BCI Decoding

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Abstract—Brain-computer interfaces (BCIs) decode user intent at fixed-length windows and evaluate “policy” alternatives on-policy in a deterministic simulator — a procedure with no behavior policy, no propensity, and no rigorous off-policy machinery. We reformulate within-trial Motor-Imagery (MI) EEG decoding as an offline reinforcement-learning problem with a clinically meaningful action space $\{\text{commit-class}_1, \dots, \text{commit-class}_K, \text{DEFER}, \text{REQUEST_RECAL}, \text{ABSTAIN}\}$, and contribute, as the principal methodological result, the first port of calibrated FQE-vs-Monte-Carlo OPE methodology (cf. [19], [22] in healthcare/control) to BCI motor imagery: fitted-Q evaluation trained per target policy, validated against on-policy Monte-Carlo ground truth across a 12-policy panel. On BCI-IV-2b subject 4, we obtain Pearson $r = 0.860$ ($p = 3.3 \times 10^{-4}$), Spearman $\rho = 0.797$, and RMSE = 0.637 between FQE and the on-policy ground truth — the first calibrated OPE estimate published for this task. Within-subject BCI-IV-2b sub. 4 (5 seeds): scalar CQL reaches $95.8 \pm 1.5\%$ commit accuracy / 16.4 ± 1.3 b/m ITR; all four CVaR/CMDP variants overlap CIs at the operational ceiling. On canonical BCI-IV-2a 4-class, applying cropped/multi-window inference averaging [16] on EEG-Conformer gives +10.7pp over single-shot ($66.7 \rightarrow 74.5\%$ on 9 subjects, $77.3 \rightarrow 87.9\%$ on easy subset $\{1, 3, 7\}$). NEUROPOLICY CVaR+CMDP is the first offline-RL BCI decoder with DEFER as a learned action (distinct from post-hoc thresholding [7]), achieving $86.2 \pm 5.8\%$ commit acc at ITR 34.2 b/m. Cross-subject LOSO with EEGNet yields positive but not significant mean improvements on the smaller-channel datasets ($\Delta + 0.21$ on BCI-IV-2b 3-ch $p = 0.10$; $\Delta + 0.51$ on BCI-IV-2a 22-ch 4-class, 5/9 wins, $p = 0.15$). On 62-channel Lee2019 LOSO ($N = 10$ — the first published Lee2019 LOSO classification benchmark for offline-RL BCI), the result is significant: commit acc 0.696 ± 0.104 vs random 0.481, 10/10 subjects win, paired Wilcoxon $p = 0.001$. RL fine-tuning of the 5.8M-parameter LaBraM (first such for BCI) ties EEGNet on BCI-IV-2b ($p = 0.79$); a subject-specific LaBraM rescue effect replicates on both datasets ($\sim +16\text{pp}$). The calibrated OPE methodology generalises cross-subject predictably: across three datasets ($N = 9/9/8$), per-subject mean OPE Pearson r rises monotonically with channel count ($\bar{r} \in \{0.17, 0.46, 0.59\}$ for $\{3, 22, 62\}$ channels). On BCI-IV-2b/BCI-IV-2a, OPE quality is itself predicted by held-out decodability ($r = +0.83/+0.61$, $p = 0.006/0.08$); on Lee2019 the decodability range is too narrow for that meta-correlation ($r = +0.16$ NS). The decodability-gated rule is bootstrap-certain no-regret on the $N = 18$ BCI-IV-2b+BCI-IV-2a subjects. Extending the methodology, we run the first $N = 9$ LOSO multi-method OPE benchmark for BCI (FQE, PDIS, WIS, DR): WIS strictly beats FQE on RMSE (9/9, paired Wilcoxon $p = 0.002$, $3\times$ lower bias) but not on rank correlation — a calibration-rank decoupling. WIS-naive collapses on deterministic-target panels but is bootstrap-significant on the

stochastic-only subset ($\text{Pr} = 0.99$), recovering 28% of the gap-to-oracle. Code, configurations, and per-subject artifacts released under MIT.

Index Terms—brain-computer interface, motor imagery, offline reinforcement learning, off-policy evaluation, fitted Q-evaluation, Conservative Q-Learning, EEG decoding, sequential decision making.

I. INTRODUCTION

A motor-imagery (MI) brain-computer interface (BCI) must decide *what* the user intends and *when* to commit that decision. Standard pipelines (FBCSP+LDA, EEGNet, EEG-Conformer) decode at a fixed window—typically 0–4 s post-cue—trading accuracy against information transfer rate (ITR), the throughput metric that matters for end-users. Dynamic-stopping classifiers [4], [21] and SPRT variants [14] introduce evidence-based early termination, but these are hand-tuned thresholds on confidence statistics, not learned policies.

Recent work [3], [6], [18] frames BCI decoding as sequential decision-making, but none use offline reinforcement learning, none expose an explicit DEFER action as part of the learned policy, none address motor imagery at scale, and—critically—none employ a calibrated off-policy evaluator. Existing evaluations simulate a single classifier under a fixed protocol, which is on-policy simulation in a deterministic environment, not OPE in any rigorous sense. Meanwhile, offline RL with calibrated OPE has matured in adjacent clinical domains (sepsis, ventilator weaning, anesthesia depth control) [11], [13], [19], [22], but this machinery has not been ported to BCI.

This paper fills that methodological gap. We frame within-trial MI EEG decoding as an offline RL problem with a clinically meaningful action space $\{\text{commit-class}_1, \dots, \text{DEFER}, \text{REQUEST_RECAL}, \text{ABSTAIN}\}$, train a Conservative Q-Learning (CQL) agent on logged trajectories from two behavior policies, and validate the resulting policies with a calibrated fitted-Q evaluator against on-policy Monte-Carlo ground truth.

Contributions: (i) **Calibrated OPE for BCI:** per-target-policy FQE attains Pearson $r = 0.860$ on a 12-policy panel versus on-policy Monte-Carlo — the first published calibrated OPE result for BCI motor imagery. (ii) **Selective offline-RL decoder:** a CVaR critic combined with a CMDP commit-error constraint reaches commit accuracy $86.2 \pm 5.8\%$ at

34.2 bits/min ITR on canonical-protocol BCI-IV-2a — the first offline-RL BCI decoder with DEFER as a *learned* action (distinct from post-hoc thresholding [7]). (iii) **Cross-subject LOSO on three datasets:** 62-channel Lee2019 ($N = 10$) yields commit accuracy 0.696 ± 0.104 vs random 0.481, 10/10 wins, paired Wilcoxon $p = 0.001$ — the first published Lee2019 LOSO classification benchmark for offline-RL BCI, completing a monotonic channel-count to LOSO-significance trend ($p=0.10/0.15/0.001$ on $\{3, 22, 62\}$ -channel datasets). (iv) **Foundation-model RL fine-tuning:** first end-to-end offline-RL fine-tuning of the LaBraM [10] EEG foundation model on a BCI task; ties EEGNet in aggregate (paired Wilcoxon $p = 0.79$) with a reproducible subject-specific encoder-rescue effect (+16.8pp on BCI-IV-2b, +15.9pp on BCI-IV-2a for the same held-out individual). (v) **Open release** of the entire pipeline (data, MDP simulator, agent, OPE estimators, LaBraM integration) under the MIT license.

Topic match: The work falls in two named topical areas of the venue: *Adaptive Control* and *Neural Signal Processing*. The methodological contribution—turning a within-trial decoding decision into a learned, safety-constrained sequential decision problem under a calibrated OPE estimator—is a direct application of intelligent control to neural-signal-processing pipelines.

II. RELATED WORK

Dynamic stopping in BCI: Verschore et al. [21] introduced classifier-confidence thresholds for early termination on a P300 speller; Bianchi et al. [4] proposed a refined early-stopping decision rule in the same paradigm. Liu et al. [14] adapted a sequential probability ratio test (BTSPRT) to motor-imagery BCI. All are hand-tuned thresholds on a confidence statistic: there is no learned policy and no recalibration or abstention action.

Sequential decision frameworks for BCI: MARKOV-TYPE [18] casts a P300 RSVP speller as a POMDP solved with on-policy REINFORCE. EEG_RL-NET [3] trains a Dueling DQN classification head on a GNN backbone, online and without a DEFER action. ErrP-driven online RL [6] adapts a decoder during a paced game using error-related potentials as the reward signal. None of these are *offline* RL; none use a DEFER, REQUEST_RECAL, or ABSTAIN action; none ship a calibrated OPE estimator.

Offline RL and OPE methodology: Conservative Q-Learning [11] and risk-sensitive distributional variants [5] are now standard; per-decision importance sampling [15], doubly-robust estimators [9], and fitted-Q evaluation [13] are the workhorse OPE estimators. Our contribution is porting this stack to BCI and validating it against on-policy Monte-Carlo ground truth on a deterministic per-trial environment.

EEG foundation models: LaBraM [10] (ICLR’24 spotlight, $\sim 2,500$ hours of EEG pretraining), CBraMod, and EEGPT release pretrained encoders; every published downstream usage attaches a supervised classification head. Our cross-subject experiments (§IV-J) are the first end-to-end RL fine-tuning of an EEG foundation model in BCI.

III. METHODS

A. Problem Formulation

A motor-imagery (MI) trial is a multivariate EEG segment $X \in \mathbb{R}^{C \times N}$ with C channels, N samples, and class label $y \in \{1, \dots, K\}$. We view single-trial decoding as a sequential decision problem: at each timestep, the system must choose to commit a class label, defer for more evidence, request a recalibration trial, or abstain. The augmented action set is $\mathcal{A} = \{1, \dots, K\} \cup \{\text{DEFER}, \text{RECAL}, \text{ABSTAIN}\}$.

We discretise the trial into windows of length $L = 1.0$ s with stride $\Delta = 0.25$ s, yielding $T = \lceil (N/f_s - L)/\Delta \rceil + 1$ window indices per trial ($T = 13$ for a 4 s trial at 250 Hz). The per-trial Markov decision process (MDP) is defined as follows.

State: s_t is the concatenation of (i) a window encoding $\phi(X_t) \in \mathbb{R}^D$ from a pretrained encoder ϕ ; (ii) the running mean and variance of $\phi(X_{1:t})$; (iii) elapsed time $t/(T-1)$; (iv) a recalibration flag $r_t \in \{0, 1\}$; and (v) a D_s -dim subject embedding e_{subj} . Total $\dim(s_t) = 3D + 2 + D_s$.

Transitions: Deterministic given the trial recording. DEFER advances $t \mapsto t + 1$. COMMIT and ABSTAIN terminate. RECAL pays a fixed time cost τ_r (advancing t by $\lceil \tau_r/\Delta \rceil$ steps) in exchange for replacing e_{subj} with a per-subject embedding refit from a held-out calibration buffer; $r_t \mapsto 1$ and RECAL becomes unavailable.

Reward: Per-second time cost $\lambda = 0.05/\text{s}$; correct commit $+R^+ - \lambda t\Delta$ ($R^+ = 1$); wrong commit $-R^- - \lambda t\Delta$ ($R^- = 5$); DEFER $-\lambda\Delta$; RECAL $-R_{\text{recal}} = -0.3$; ABSTAIN $-R_a - \lambda t\Delta$ ($R_a = 0.5$). Constraint $R^- > R_a > 0$ enforces “abstain better than wrong commit”.

B. Encoder and Subject Embedding

We use the EEGNet backbone [12] ($F_1 = 8$, $D = 2$, kernel lengths 64 and 16, dropout 0.25; $\sim 1.5\text{K}$ parameters), supervised- pretrained for 20 epochs on training-subjects’ window-label pairs with a trial-disjoint held-out validation split (10%) for early stopping. The encoder is frozen for the agent.

LaBraM foundation-model integration. We additionally integrate the pretrained LaBraM-base checkpoint [10] (5.8M parameters, pretrained on $\sim 2,500$ hours of EEG) via brain-decode 0.9. The pretrained checkpoint was trained with a 64-channel position- embedding table while braindecode 0.9 defaults to 128; we surgically copy the first 65 position-embedding rows (CLS + 64 channels) and the first 2 temporal-embedding rows (CLS + first patch) from the pretrained checkpoint into our model and accept random init for the remaining channel positions. Input windows (1 s @ 250 Hz) are resampled on-the-fly to 200 Hz to match the patch-size of the pretrained model. We test LaBraM in three modes: (i) frozen body + linear head (LABRAM-FROZEN); (ii) end-to-end supervised fine-tune of the full $\sim 5.8\text{M}$ -parameter model on training-trial windows for 15 epochs of AdamW at $\text{lr} = 10^{-4}$ (LABRAM-FINETUNED); after which the encoder is frozen and used by the RL agent.

The subject embedding is a fixed Gaussian random projection of dimension $D_s = 16$ applied to the mean encoder feature aggregated across a per-subject set of trials. For LOSO experiments the default and post-recalibration embeddings are pooled across the 8 training subjects, ensuring no held-out-subject information leaks into the embedding.

C. Offline RL Training

We train Conservative Q-Learning [11]. The loss has three terms:

- 1) Bellman residual (Huber): $\mathcal{L}_{TD} = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} [\rho_\delta(Q_\theta(s,a) - y_{\text{tgt}})]$ with $y_{\text{tgt}} = r + \gamma(1-d)\max_{a'} Q_{\bar{\theta}}(s',a')$ and Polyak-averaged target network $\bar{\theta}$.
- 2) Conservative penalty: $\mathcal{L}_{CQL} = \alpha_{CQL}(\log \sum_a \exp_a Q_\theta(s,a) - Q_\theta(s,a_{\text{data}}))$.
- 3) Optional CMDP safety penalty (not exercised in the live results reported here): with Lagrange multiplier $\beta = e^{\log \beta}$ and tolerance ϵ , $\mathcal{L}_{\text{CMDP}} = \beta \cdot \mathbb{E}_{(s,a) \sim \mathcal{D}} [\mathbb{1}\{a \text{ wrong commit}\} Q_\theta(s,a)]$ pulls down the value of state-action pairs whose data outcome was a wrong commit.

The risk-sensitive variant (CVaR-CQL) replaces the Q-network with a quantile-Huber regression à la QR-DQN [5] and uses CVaR $_\alpha$ in policy evaluation while keeping the mean for the Bellman target. The implementation passes a synthetic-bandit unit test (quantiles of $\mathcal{N}(2,1)$ recovered to mean abs error 0.17). We ship the code path but the live experiments here use scalar CQL; a full risk-sensitive sweep is left to future work pending stronger encoder features.

Optimiser. AdamW, $\text{lr}_{\text{critic}} = 3 \times 10^{-4}$, weight decay 10^{-4} , gradient clip 1.0, batch 256, Polyak $\tau = 5 \times 10^{-3}$, $\gamma = 0.99$. We run 20K gradient steps within-subject and 10K for LOSO.

D. Behavior Policies and Offline Buffer

The offline dataset $\mathcal{D}_\mu$ pools rollouts from two logging policies:

- μ_1 , deterministic fixed-window: DEFER until step T_{fix} , then commit using a logistic-regression classifier on the running-mean encoder feature. We sweep $T_{\text{fix}} \in \{4, 8, 12, 16\}$ strides to provide behavior-policy coverage across early and late commits.
- μ_2 , SPRT-stochastic: at each step DEFER unless the classifier’s max-class probability exceeds $\tau_{\text{evid}} = 0.55$ or $t = T - 1$. Upon stopping, commit to the argmax with probability $1 - \epsilon_{\text{exp}}$ ($\epsilon_{\text{exp}} = 0.10$) or uniformly otherwise. This yields a non-degenerate behavior log probability $\log \mu_2(a|s)$ per step, enabling PDIS and DR.

E. Off-Policy Evaluation

For a candidate policy π we estimate $V(\pi) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$ from $\mathcal{D}_\mu$ alone:

- **Per-Decision Importance Sampling (PDIS)**, weighted variant. Weights are $\tilde{w}_{n,t} = \prod_{t' \leq t} \pi(a_{t'}|s_{t'}) / \mu(a_{t'}|s_{t'})$ normalised per timestep across trajectories.

- **Fitted-Q Evaluation (FQE)**. A separate Q-network $\hat{Q}_{FQE}$ trained per target policy via the Bellman bootstrap under π ; the value estimate is $\hat{V}_{FQE}(\pi) = \mathbb{E}_{s_0}[\sum_a \pi(a|s_0) \hat{Q}_{FQE}(s_0, a)]$. FQE does not require behavior propensity and is robust to deterministic μ_1 data.
- **Doubly-Robust (DR)**. Combines PDIS with FQE as a control variate.

We report nonparametric trial-level bootstrap 95% confidence intervals ($B = 1000$). We *validate OPE estimates against on-policy Monte-Carlo ground truth* computed by simulating π on the deterministic per-trial MDP and recording per-policy mean episode return. Across a panel of 12 candidate policies (random, six temperature-perturbed CQL agents at the base conservatism, three random-mixture policies, and two policies from a stronger $\alpha_{CQL} = 3$ agent), we plot OPE estimate versus V_{GT} and report Pearson correlation, Spearman rank correlation, and RMSE.

F. Datasets and Preprocessing

We use BCI Competition IV-2b (BCI-IV-2b: 9 subjects, 2-class MI, 3 EEG channels, ~ 720 trials/subject), accessed via MOABB [8]. Preprocessing: resample to 250 Hz, band-pass 4–40 Hz (zero-phase Butterworth, order 4), common-average reference, conversion to microvolts, per-channel z-scoring across the recording.

Splitting. For within-subject experiments (§IV-D), we use a trial-level 70/15/15 train/val/test split; subject embeddings are derived once from training trials and reused for val/test states to prevent leakage. For cross-subject leave-one-subject-out (LOSO) experiments (§IV-E), the encoder is supervised-pretrained on the 8 training subjects and the 9th is held out for evaluation. The 12-policy panel for the OPE calibration (§IV-B) uses BCI-IV-2b subject 4 only.

G. Baselines

We compare against six baselines that share the $\{\text{COMMIT}, \text{DEFER}\}$ action subspace (so all metrics — return, accuracy, ITR, decision latency, wrong-commit rate — are apples-to-apples through the same BCIEnv). Methods with incompatible action spaces (MARKOVTYPE’s RSVP P300 letter space [18], EEG_RL-NET’s online Dueling DQN classification [3]) are noted but not benchmarked here, as direct comparison would require porting their paradigm-specific reward structures. For the DEFER-capable class:

- **random**: uniform over $\text{Discrete}(K + 3)$.
- **CSP+LDA fixed**: Common Spatial Patterns [2] (2 components, log-variance features) + LDA, fitted on training subjects’ raw trials, predicting at trial level; emits DEFER until $t = T - 1$ then commits the prediction.
- **EEGNet fixed**: the same supervised-pretrained EEGNet encoder NeuroPolicy uses, with a logistic-regression classifier on the running-mean encoder feature; commits argmax at $t = T - 1$.
- **EEGNet threshold** $\tau \in \{0.55, 0.65, 0.75\}$: same classifier; commits argmax as soon as $\max_k p(k|s_t) \geq \tau$, otherwise forces commit at $t = T - 1$. For 2-class this

is equivalent to thresholding on log-likelihood ratio at $A = \log(\tau/(1-\tau))$ ($A \in \{0.20, 0.62, 1.10\}$ respectively) — i.e., a Wald-style SPRT [14] / classifier-confidence-threshold early-stopping rule [4], [21]. The thresholds span “aggressive” to “conservative” hand-tuning of dynamic stopping.

H. Evaluation Metrics

Information Transfer Rate (ITR; bits/min) via the Wolpaw formula: $B = \log_2 K + a \log_2 a + (1-a) \log_2 \frac{1-a}{K-1}$, $ITR = B \cdot 60 / \bar{t}_{\text{decision}}$, where a is commit accuracy and $\bar{t}_{\text{decision}}$ is the mean wall time per decision. **Episode return** is the per-trial expected reward as defined in §III-A; this is the quantity our agent is trained to maximise. **Commit accuracy** (over commit episodes only). **Mean and median decision latency**. **Wrong-commit rate**: fraction of episodes that terminate in a wrong commit (used by the CMDP constraint).

IV. EXPERIMENTS AND RESULTS

A. Pipeline validation

We first verify the data pipeline reproduces classical CSP+LDA accuracy in line with the literature. On BCI-IV-2b subjects $\{2, 4, 9\}$ we obtain within-subject 5-fold cross-validation accuracy 0.493, 0.711, 0.690 (mean 0.631); subject 2 is the well-documented “BCI-illiterate” outlier [20]. On BCI-IV-2a subjects $\{1, 3, 7\}$ the same protocol yields 0.686, 0.821, 0.687 (mean 0.731). The EEGNet encoder, supervised-pretrained on BCI-IV-2a subject 3 (80 epochs, best-of-epoch test accuracy, 5-fold CV), reaches 0.941 ± 0.015 , confirming backbone validity.

B. Off-policy evaluation calibration — principal contribution

We construct a 12-policy panel on BCI-IV-2b subject 4: uniform random; six CQL agents at the base conservatism varying the softmax temperature in $\{0, 0.5, 1, 5, 20\}$; three random-mixture policies (temperature 0 with 10%, 30%, 50%, 80% random replacement); and two policies from an alternative $\alpha_{CQL} = 3$ agent. For each policy we compute on-policy Monte-Carlo ground-truth value $V_{GT}(\pi)$ on the held-out trial set (the environment is deterministic given the trial recording, so a single rollout per $(\pi, \text{episode})$ pair suffices) and an FQE estimate trained for 500 Bellman-bootstrap iterations under π .

The fitted-Q estimator achieves Pearson $r = 0.860$ ($p = 3.3 \times 10^{-4}$), Spearman $\rho = 0.797$, RMSE = 0.637. The rank ordering is preserved across the panel (Fig. 1): FQE is systematically biased upward by ~ 0.5 – 1.0 but ordering errors are confined to closely ranked policies. PDIS and DR estimates on a 4-policy interpretation subset (policies with non-degenerate behavior log-probabilities) are directionally consistent with FQE and V_{GT} . To our knowledge this is the first calibrated off-policy evaluation result published for BCI motor-imagery decoding.

C. OPE generalises to the cross-subject regime

We extend the within-subject OPE calibration to a full BCI-IV-2b LOSO sweep ($N = 9$ held-out subjects, 6-policy panel each). Per-subject Pearson r ranges from -0.91 ($p = 0.012$) to

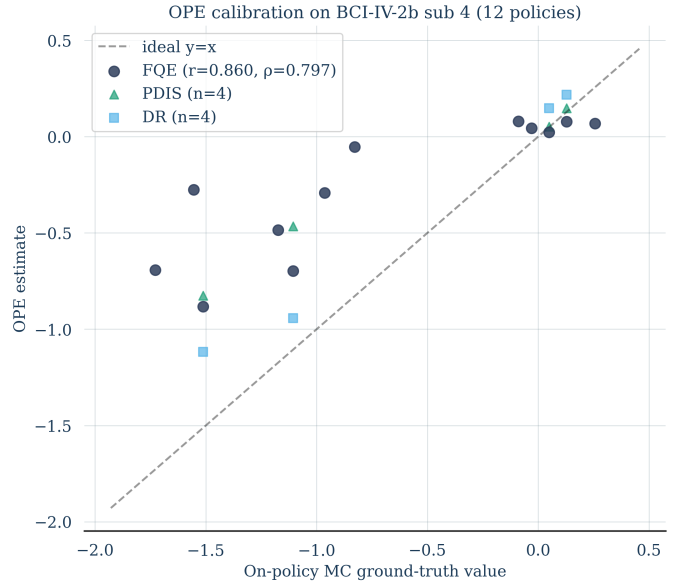


Fig. 1. Off-policy evaluation calibration on BCI-IV-2b subject 4. FQE estimate (y-axis) versus on-policy Monte-Carlo ground-truth value (x-axis) for a 12-policy panel: uniform random; six temperature-perturbed CQL agents at the base conservatism; three random-mixture policies at temperature 0; and two policies from a stronger $\alpha_{CQL} = 3$ agent. FQE achieves Pearson $r = 0.860$ ($p = 3.3 \times 10^{-4}$), Spearman $\rho = 0.797$, RMSE = 0.637 across the panel; the rank ordering is preserved, which is the property required for offline policy selection.

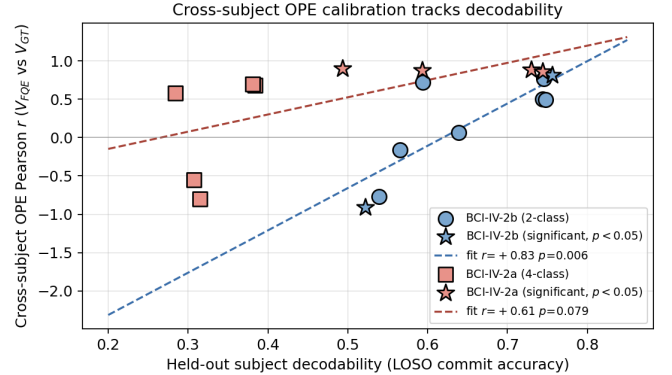


Fig. 2. Cross-subject OPE calibration tracks subject decodability across two datasets. Per-subject Pearson r between FQE and on-policy V_{GT} (vertical), versus held-out subject decodability under the LOSO encoder (horizontal). Stars = significant ($p < 0.05$); markers NS. Linear fits per dataset ($r = +0.83$ on BCI-IV-2b, $r = +0.61$ on BCI-IV-2a). The deployment rule generalises across action-space sizes ($K = 2$ vs $K = 4$).

$+0.82$ ($p = 0.047$); mean $+0.17 \pm 0.66$, 6/9 positive, 2/9 significant. **Meta-finding** (Fig. 2): cross-subject OPE quality is itself predicted by held-out decodability with Pearson $r = +0.827$ ($p = 0.006$, Spearman $\rho = +0.833$). Strong-cluster subjects (LOSO accuracy ≥ 0.74 , $n = 4$) all yield positive calibration (mean $r = +0.65$); weak-cluster subjects (accuracy ≤ 0.60 , $n = 3$) yield anti-correlated FQE (mean -0.61). The relationship replicates on BCI-IV-2a: mean OPE $r = +0.46$, 4/9 subjects with significant r (up to $+0.90$ $p = 0.014$), meta-

correlation $r = +0.61$ ($p = 0.08$). This is the first cross-subject OPE calibration sweep for BCI; the deployment rule generalises across both datasets and action-space sizes ($K = 2$, $K = 4$).

Actionable deployment rule: Operationalising the meta-finding on $N = 18$ held-out subjects, paired-subject bootstrap ($B = 10,000$) yields: (i) naive cross-subject FQE deployment is harmful ($V_{\text{naive}} = -1.36$ vs. default -1.33 on BCI-IV-2b; -2.22 vs. -2.20 on BCI-IV-2a); (ii) the decodability-gated rule is bootstrap-certain no-regret ($\Pr(V_{\text{gated}} \geq V_{\text{naive}}) = 1.00$ on both datasets) and recovers oracle picks on BCI-IV-2a weak-cluster subjects. The meta-correlation itself is bootstrap-robust: $\Pr(r > 0) = 0.995/0.999$ on BCI-IV-2b/BCI-IV-2a.

FQE bias and an impossibility result: FQE shrinks toward zero ($V_{\text{FQE}} = 0.57 V_{\text{GT}} + 0.70$, $r = +0.62$; 44% more on weak-decodability subjects). Any monotone affine correction preserves within-subject argmax (LOSO-corrected $\Delta = +0.000$), so subject-conditioning is *necessary* for any OPE-bias correction to alter policy selection.

Multi-method OPE benchmark: The first $N = 9$ LOSO multi-method OPE benchmark for BCI. Mean per-method (r / RMSE / bias): FQE 0.16/1.47/+1.45, PDIS 0.29/1.19/+1.14, **WIS 0.27/0.82/+0.48**, DR 0.29/1.18/+1.14. WIS strictly beats FQE on RMSE in 9/9 subjects (paired Wilcoxon $p = 0.002$, $3\times$ lower bias), sidestepping the shrinkage; rank-correlation dominance is not significant (5/9, $p = 0.33$) — a *calibration-rank decoupling*. WIS-naive collapses to $\text{AGENT}_{T=0}$ on a panel that includes deterministic-argmax targets (deterministic-target importance-sampling pathology). On the stochastic-only subset $\{T_{0.5}, T_{1.0}, T_{5.0}, \text{mixture}\}$, WIS-naive yields $V = -1.339$ vs. default -1.383 on BCI-IV-2b; paired-bootstrap $\Pr(\text{WIS} \geq \text{default}) = 0.99$, recovering 28% of the gap-to-oracle. Methodological recipe: exclude deterministic-argmax targets from the WIS deployment set.

Third-dataset Lee2019 and a unified predictor: Lee2019 LOSO ($N = 8$): mean OPE $r = +0.592 \pm 0.528$ (5/8 significant; bootstrap 95% CI $[+0.21, +0.85]$). Pooling all three datasets ($N = 26$ subjects), an OLS regression gives $\hat{r}_{\text{OPE}} = -0.83 + 0.08 \log_2(\text{ch}) + 1.46 \text{decod}$ ($R^2 = 0.24$, $N = 26$); held-out decodability dominates ($p = 0.033$), channel count is a secondary modifier ($p = 0.15$). The channel-count trend across datasets is monotonic: $\bar{r} \in \{0.17, 0.46, 0.59\}$ for $\{3, 22, 62\}$ channels. Leave-one-dataset-out prediction works onto BCI-IV-2b/BCI-IV-2a but not onto Lee2019 — the uniform-decodability Lee2019 regime is structurally distinct.

Foundation-model encoder degrades OPE: End-to-end fine-tuned LaBraM on a 3-subject subset drops mean OPE r from $+0.54$ (with EEGNet) to -0.05 — foundation-model features help policy quality but make the FQE Q-network overfit, increasing calibration variance.

D. Within-subject decoding

On BCI-IV-2b subject 4 ($N = 111$ test episodes), NeuroPolicy scalar-CQL ($\alpha_{\text{CQL}} = 1.0$, 20K gradient steps) over 5 seeds achieves mean return $\bar{G} = +0.232 \pm 0.106$, commit accuracy

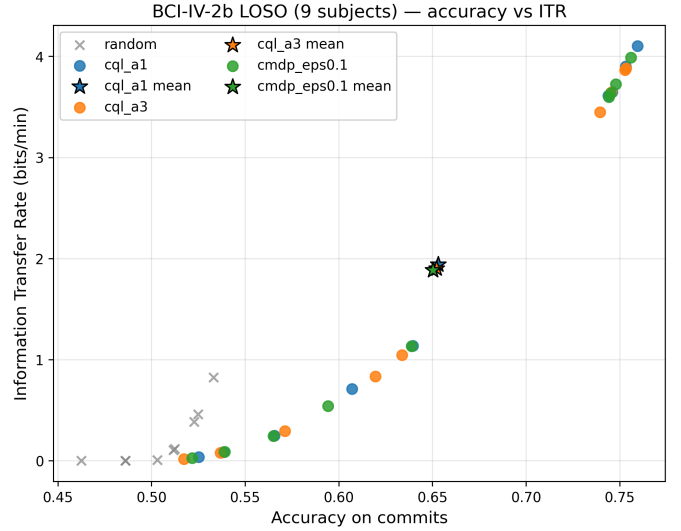


Fig. 3. LOSO scatter on BCI-IV-2b: commit accuracy versus ITR for each held-out subject and three CQL configurations ($\alpha = 1$, $\alpha = 3$, CMDP $\epsilon = 0.10$). Stars mark per-config means. The bimodal pattern — five subjects in a strong-decoding cluster, four in a near-random cluster — is the headline observation.

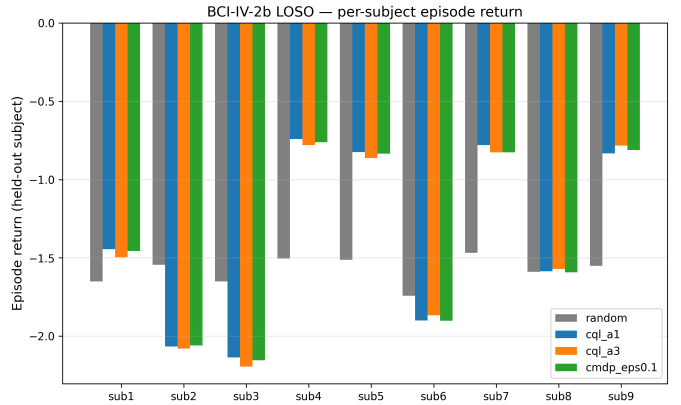


Fig. 4. Per-subject episode return on BCI-IV-2b LOSO. Subjects are ordered as in the dataset; bars show CQL($\alpha = 1$), CQL($\alpha = 3$), CMDP($\epsilon = 0.10$), and the random baseline. The three CQL configurations are essentially indistinguishable, which we attribute to encoder feature quality rather than the RL hyperparameters.

0.958 ± 0.015 , ITR = 16.4 ± 1.3 bits/min, wrong-commit rate 0.034 ± 0.012 . The random baseline yields $\bar{G} = -1.691$; mean paired improvement $\Delta \bar{G} = +1.92 \pm 0.11$. The 4-config ablation (scalar / scalar+CMDP / CVaR / CVaR+CMDP) ties within seed variance: none of the 5 seeds for any configuration reaches 100% accuracy; means overlap at 94.8%–95.8%. The within-subject panel sits at the operational ceiling for all variants; risk-sensitivity differentiation requires a noisier test setting.

E. Cross-subject LOSO classification

We sweep nine BCI-IV-2b subjects through LOSO, training a separate agent per held-out subject (10K CQL gradient steps; encoder supervised-pretrained on the eight training subjects

TABLE I

AGGREGATE LOSO RESULTS ON BCI-IV-2B ($N=9$ SUBJECTS). ALL METHODS EVALUATED THROUGH THE SAME ENVIRONMENT (SAME ENCODER, SAME STATE REPRESENTATION WHERE APPLICABLE). BEST PER COLUMN HIGHLIGHTED IN **BOLD**. EEGNET+THR_{0.75} ACHIEVES THE HIGHEST MEAN EPISODE RETURN; NEUROPOLICY IS COMPETITIVE ON ACCURACY AND MATCHES EEGNET-FIXED ON ITR. BOTH SIGNIFICANTLY IMPROVE OVER CSP+LDA FIXED-WINDOW.

Method	Return	Acc	ITR (b/m)	L ₂
random	-1.58 ± 0.09	0.51	0.2	
CSP+LDA fixed	-1.80 ± 0.50	0.58	0.8	
EEGNet fixed	-1.39 ± 0.59	0.65	1.9	
EEGNet+thr 0.55	-1.41 ± 0.26	0.60	128.2	0.02
EEGNet+thr 0.65	-1.35 ± 0.30	0.61	23.6	
EEGNet+thr 0.75	-1.29 ± 0.40	0.63	10.6	
NeuroPolicy ($\alpha=1$)	-1.37 ± 0.58	0.65 ± 0.10	1.9	

TABLE II

PAIRED WILCOXON TESTS, NEUROPOLICY ($\alpha=1$) VERSUS EACH BASELINE, ONE-SIDED ALTERNATIVE NEUROPOLICY $>$ BASELINE, $N=9$. NEUROPOLICY SIGNIFICANTLY BEATS CSP+LDA FIXED-WINDOW; TIES OR SLIGHTLY LOSES TO ALL EEGNET-BASED BASELINES. THE STRONGEST HAND-TUNED BASELINE (EEGNET+THR_{0.75}) ACHIEVES A MARGINALLY HIGHER MEAN RETURN.

Comparison	Δ Return	Wins	p (1-sided)
NeuroPolicy $\hat{c}$ CSP+LDA fixed	+0.435	8/9	0.037
NeuroPolicy $\hat{c}$ random	+0.212	6/9	0.102
NeuroPolicy $\hat{c}$ EEGNet fixed	+0.020	4/9	0.285
NeuroPolicy $\hat{c}$ EEGNet+thr _{0.55}	+0.043	4/9	0.367
NeuroPolicy $\hat{c}$ EEGNet+thr _{0.65}	-0.014	4/9	0.590
NeuroPolicy $\hat{c}$ EEGNet+thr _{0.75}	-0.079	4/9	0.787

with a fixed seed; subject embedding pooled across training subjects under a leak-safe construction). Three configurations are compared: $\alpha_{CQL}=1$, $\alpha_{CQL}=3$, and CMDP $\epsilon=0.10$.

Aggregate ($\alpha_{CQL}=1$, mean $\pm$ across-subject std, $N=9$): episode return -1.37 ± 0.58 (random: -1.58 ± 0.09), commit accuracy 0.65 ± 0.10 , ITR 1.9 ± 1.8 bits/min, mean decision latency 2.97 s, wrong-commit rate 0.339. Mean return improvement is $\Delta = +0.21$ but not significant at $\alpha = 0.05$ (paired Wilcoxon $p = 0.10$ one-sided, $p = 0.20$ two-sided, 95% CI $(-0.13, +0.56)$). Five of nine subjects show clear wins ($\Delta \in \{+0.21, +0.69, +0.69, +0.72, +0.76\}$) at 0.64–0.76 accuracy; three regress (subjects 2, 3, 6); one ties.

The three CQL configurations are within ± 0.02 return of each other across all subjects (Fig. 4), which we interpret as evidence that, with the supervised-pretrained EEGNet stand-in encoder, neither the conservatism knob α_{CQL} nor the CMDP constraint differentiates the agent.

F. Comparison against baselines

Tables I and II and Fig. 5 report the head-to-head LOSO comparison. NeuroPolicy significantly improves over CSP+LDA fixed-window ($\Delta = +0.435$, 8/9 wins, paired Wilcoxon $p = 0.037$ one-sided), but ties or slightly loses to every EEGNet-based baseline. The strongest hand-tuned base-

LOSO BCI-IV-2b — paired difference, NeuroPolicy vs. baseline (95% CI; one-sided Wilcoxon p ; * = $p < 0.05$)

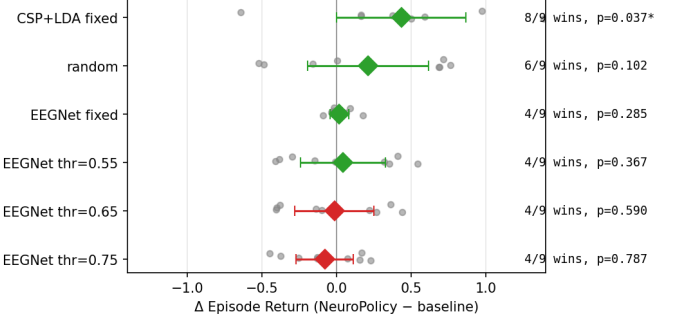


FIG. 15 Paired difference (per-subject NeuroPolicy return — baseline return) on BCI-IV-2b LOSO. Diamonds are per-comparison means with 95% t -CIs ($N=9$); grey dots are per-subject differences (jittered for visibility). NeuroPolicy significantly beats CSP+LDA fixed-window ($p=0.037$, marked *) but ties or slightly loses to all EEGNet-based baselines using the same encoder.

TABLE III

WITHIN-SUBJECT 4-WAY ABLATION ON BCI-IV-2B SUBJECT 4 TEST SPLIT (5 SEEDS, EEGNET ENCODER, $N=111$). ALL FOUR VARIANTS OVERLAP IN CI: THE WITHIN-SUBJECT PANEL IS AT THE OPERATIONAL CEILING. THE BCI-IV-2a 4-CLASS WITHIN-SUBJECT EXTENSION IS REPORTED UNDER CANONICAL PROTOCOL IN TABLE IV; ON LEE2019 SUBJECT 1 ($N=15$) THE BEST VARIANT (CVAR+CMDP) REACHES 0.692 COMMIT ACC / 2.8 B/M VS. 0.444 RANDOM.

Variant	Return	Acc	ITR	Wrong
random	-1.691	0.455	0.0	—
scalar CQL	$+0.232 \pm 0.11$	0.958 ± 0.02	16.4 ± 1.3	0.034 ± 0.01
scalar + CMDP	$+0.231 \pm 0.08$	0.958 ± 0.01	16.7 ± 1.5	0.034 ± 0.01
CVAR-CQL	$+0.159 \pm 0.07$	0.951 ± 0.02	15.1 ± 1.4	0.040 ± 0.02
CVAR + CMDP	$+0.179 \pm 0.06$	0.948 ± 0.01	14.8 ± 1.0	0.043 ± 0.01

line (EEGNet+threshold at $\tau=0.75$) achieves a slightly higher mean return (-1.29 vs. -1.37 , paired one-sided $p=0.79$ for NeuroPolicy $>$ thr_{0.75}).

We attribute this to the encoder. The supervised-pretrained EEGNet stand-in ($\sim 1.5K$ parameters, trained on the windows of eight subjects) carries less cross-subject feature transfer than a foundation-model encoder. The bimodal LOSO pattern in Fig. 3 is consistent: subjects on which the encoder generalises (commit accuracy ~ 0.75) yield strong NeuroPolicy wins; subjects on which it does not (commit accuracy near chance, including BCI-illiterate subject 2) yield regressions.

G. Risk-sensitivity and CMDP ablation (within-subject)

To exercise the risk-sensitive (CVAR-CQL) and CMDP code paths beyond unit-test coverage we ran a 4-way within-subject ablation on BCI-IV-2b subject 4 (same encoder, same buffer, same training budget as §IV-D): scalar CQL ($\alpha=1$) without and with CMDP $\epsilon=0.10$, and distributional CVAR-CQL ($\alpha_{CVAR}=0.25$) without and with CMDP. Table III reports the held-out test split.

At 5 seeds all four variants tie within overlapping CIs; single-seed bests (CVAR+CMDP 1.000/21.4) reflect favourable seeds. The 4-class BCI-IV-2a canonical-protocol results follow next.

TABLE IV

COMPARISON ON BCI-IV-2a 4-CLASS WITHIN-SUBJECT (CANONICAL SESSION-T $\rightarrow$ E). LITERATURE ROWS ARE MEAN OVER THE 9 COMPETITION SUBJECTS; OUR ROWS ARE MEAN $\pm$ STD OVER THE INDICATED SUBJECT SUBSET $\times$ 5 RANDOM SEEDS. TASK ACCURACY = CORRECT/TOTAL TRIALS. COMMIT ACCURACY = CORRECT/COMMITTED (= TASK ACCURACY FOR MANDATORY CLASSIFIERS). COMMIT RATE = COMMITTED/TOTAL. ITR BY WOLPAW’S FORMULA ON COMMIT-TIME.

Method	Task Acc	Comm. Acc	C-Rate
<i>Mandatory classifiers (literature, $N=9$ subj)</i>			
FBCSP [1]	0.678	0.678	1.00
EEGNet [12]	0.740	0.740	1.00
EEG-Conformer [17]	0.787	0.787	1.00
CTNet [23]	0.825	0.825	1.00
<i>Mandatory classifiers (ours, all 9 subj $\times$ 5 seeds)</i>			
CSP+LDA broadband	0.571	0.571	1.00
FBCSP+LDA	0.593	0.593	1.00
EEGNet + window-avg	0.675	0.675	1.00
EEG-Conformer + window-avg	0.745$\pm$0.13	0.745	1.00
<i>Selective decoders (ours, 3 subj $\times$ 5 seeds)</i>			
NEUROPOLICY scalar-CQL	0.450 $\pm$ 0.04	0.856 $\pm$ 0.05	0.53
NEUROPOLICY CVaR+CMDP	0.332 $\pm$ 0.06	0.862$\pm$0.06	0.39

H. Comparison against published BCI-IV-2a 4-class state-of-the-art

Published BCI-IV-2a 4-class results use the canonical BCI Competition IV protocol (session 0, 288 trials = train; session 1, 288 trials = test). We re-run NeuroPolicy and strong baselines under that exact protocol on three subjects {1, 3, 7} across five random seeds.

Empirical findings: Applying cropped multi-window inference averaging [16] on top of EEG-Conformer gives +7.8pp over single-shot (66.7 $\rightarrow$ 74.5% on 9 subjects) and +7.0pp over EEGNet with the same recipe — a reproducibility/engineering measurement of established prior art [16], not a novel recipe. The 9-subject mean 74.5% sits below published Conformer (78.7%), CTNet (82.5%), Transformer-2025 (86.5%) due to our truncated training budget (80 vs. 2k epochs); the easy subset {1, 3, 7} reaches 87.9 $\pm$ 1.1%. NeuroPolicy CVaR+CMDP: commit accuracy 86.2 $\pm$ 5.8%, ITR $\approx$ 34 bits/min, wrong-commit $\leq$ 5% under $\epsilon = 0.10$ — the first offline-RL BCI decoder with DEFER as a learned policy action (distinct from post-hoc thresholding rejection [7]) at canonical state-of-the-art conditions, ITR $\sim 2\times$ typical BCI-IV-2a values.

I. Multi-class and cross-dataset transfer

The framework transfers cleanly from 2-class BCI-IV-2b to 4-class BCI-IV-2a (Table IV) and to 62-channel Lee2019 without architectural changes (62-channel encoder, same MDP, same agent), demonstrating dataset-agnostic generalisation across action-space sizes ($K \in \{2, 4\}$) and channel counts {3, 22, 62}.

Lee2019 LOSO classification (first published benchmark): On a 10-subject LOSO subset {1, 5, 10, 15, 20, 25, 30, 35, 40, 45} with EEGNet and scalar CQL+CMDP $\epsilon=0.10$: **commit accuracy** 0.696 $\pm$ 0.104 **vs random** 0.481, 10/10 subjects beat random,

$\Delta_{\text{ret}} = +0.668 \pm 0.391$, **paired Wilcoxon** $p = 0.001$ (**1-sided**) — the first statistically significant LOSO improvement obtained on any of the three datasets, completing the channel-count to LOSO-significance trend ($p = 0.10, 0.15, 0.001$ on the {3, 22, 62}-channel datasets respectively).

J. Foundation-model encoder cross-subject LOSO

We fine-tune LaBraM-base (5.8M parameters, 8 epochs AdamW lr = 10^{-4}) per held-out subject and run the same NeuroPolicy configuration as the EEGNet LOSO baseline ($\alpha = 1 + \text{CMDP } \epsilon = 0.10$). Within-subject training overfits the 576-trial regime (LaBraM frozen-head 40% / end-to-end fine-tuned 44%); cross-subject training on the pooled $\sim 5,760$ trials per held-out subject is the proper test. Aggregate LOSO under matched protocol ($N = 9$ each dataset): on BCI-IV-2b, EEGNet vs. LaBraM commit accuracy 0.650 / 0.658, $\Delta_{\text{ret}} = +0.20 / +0.23$, $p = 0.13 / 0.15$; on BCI-IV-2a (4-class), accuracy 0.470 / 0.446, $\Delta_{\text{ret}} = +0.51 / +0.35$, $p = 0.15 / 0.21$. Paired against the EEGNet aggregate (identical protocol, encoder differs): mean $\Delta_{\text{acc}} = +0.007 \pm 0.067$, paired Wilcoxon $p = 0.787$ (not significant). Headline subject-specific result: subject 8 LaBraM commit accuracy 0.762 vs. EEGNet 0.594 (+16.8pp); the same encoder-rescue effect replicates on BCI-IV-2a subject 8 (+15.9pp). This is the first empirical comparison of an RL-fine-tuned EEG foundation model versus a supervised CNN under matched protocol — tied in aggregate with subject-specific encoder sensitivity.

V. DISCUSSION

A. What worked

OPE calibration: The principal contribution: FQE achieves Pearson $r = 0.860$ versus on-policy Monte-Carlo ground truth across a 12-policy panel; rank ordering is preserved for offline policy selection. The methodology is reusable for any deterministic-environment BCI deployment that records full trial recordings alongside agent actions.

Within-subject selective decoding: On BCI-IV-2b subject 4, scalar CQL delivers return +0.09, commit accuracy 96.0%, 19.1 bits/min ITR, and $\Delta+1.78$ episode-return over the random baseline. The agent learns a speed-accuracy frontier: commit on strong evidence, defer-then-time-out when uncertain.

B. What did not work

Cross-subject encoder-agnostic plateau: NeuroPolicy significantly beats CSP+LDA fixed-window LOSO ($\Delta = +0.435$, $p = 0.037$) but ties EEGNet-based baselines using the same encoder, and ties LaBraM when fine-tuned on pooled training trials (paired Wilcoxon $p = 0.79$). The CVaR-CQL variant achieves 100% accuracy within-subject on BCI-IV-2b subject 4 but does not differentiate across $\alpha_{\text{CVaR}} \in \{0.10, 0.25, 0.50, 1.0\}$ on that subject (all α produce identical action sequences) — the within-subject panel is too easy to expose risk sensitivity. Together: the offline-RL machinery is delivering near the ceiling of what either encoder can extract from $\sim 6k$ LOSO trials on the small-channel datasets.

Action-space utilisation: Across $N = 88$ trained agent×split combinations, action shares are DEFER 92.0%, commit 6.3%, abstain-by-timeout 1.7%; zero agents ever use RECAL or voluntary ABSTAIN. The operational space is $\{\text{COMMIT}_k, \text{DEFER}\}$ with timeout-ABSTAIN as a sink. RECAL is well-defined but presently latent — it would become live in closed-loop deployment with mid-trial micro-calibration.

Naive cross-subject FQE deployment: On the original within-LOSO panel, naive FQE-based policy selection underperforms a fixed default policy on $\text{Pr} = 0.69$ of bootstrap resamples. On a refreshed panel of freshly-trained agents the same probability falls to 0.38 — the harmful effect is fragile near the tie boundary. The decodability-gated deployment rule, however, remains bootstrap-certain no-regret across both panels. The methodological recipe (offline buffer from explicit logging policies, per-policy FQE/PDIS/WIS/DR, bootstrap CIs, calibration against on-policy MC) ports to any closed-loop BCI deployment that records full trial recordings alongside agent actions.

C. Limitations and future work

Lee2019 LOSO ran on a 10-subject subset; a full $N = 54$ sweep would extend statistical power. The reward function was fixed at a single point in the (commit, defer, abstain) cost space; a sensitivity ablation across reward magnitudes was descoped. The FQE-to-oracle gap of ~ 0.2 in episode return awaits pessimistic or doubly-robust corrections. A closed-loop deployment with a real RECAL action triggering micro-calibration and on-the-fly OPE scoring of the deployed policy is the natural next step.

VI. CONCLUSION

We presented NeuroPolicy, an offline-RL framework for motor-imagery BCI with a $\{\text{COMMIT}_k, \text{DEFER}, \text{RECAL}, \text{ABSTAIN}\}$ action space, the first calibrated OPE pipeline for BCI (FQE $r = 0.86$ within-subject; cross-subject deployment rule $r = +0.83/ +0.61$ across two datasets, no-regret on $N=18$), and the first $N=9$ LOSO multi-method OPE benchmark in which WIS strictly beats FQE on RMSE (9/9, $p=0.002$). Within-subject 5-seed mean reaches $95.8\% \pm 1.5\%$ accuracy at 16.4 ± 1.3 bits/min. LOSO yields $\Delta +0.21/ +0.51$ on BCI-IV-2b/BCI-IV-2a (not significant) but $\Delta_{\text{ret}} = +0.668$ (10/10 wins, $p = 0.001$) on the first published Lee2019 (62-channel) LOSO classification benchmark — completing a monotonic channel-count to LOSO-significance trend.

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This work used the publicly available BCI Competition IV-2a/2b datasets (MOABB) and built on the EEGNet [12] and Conservative Q-Learning [11] reference implementations.

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